

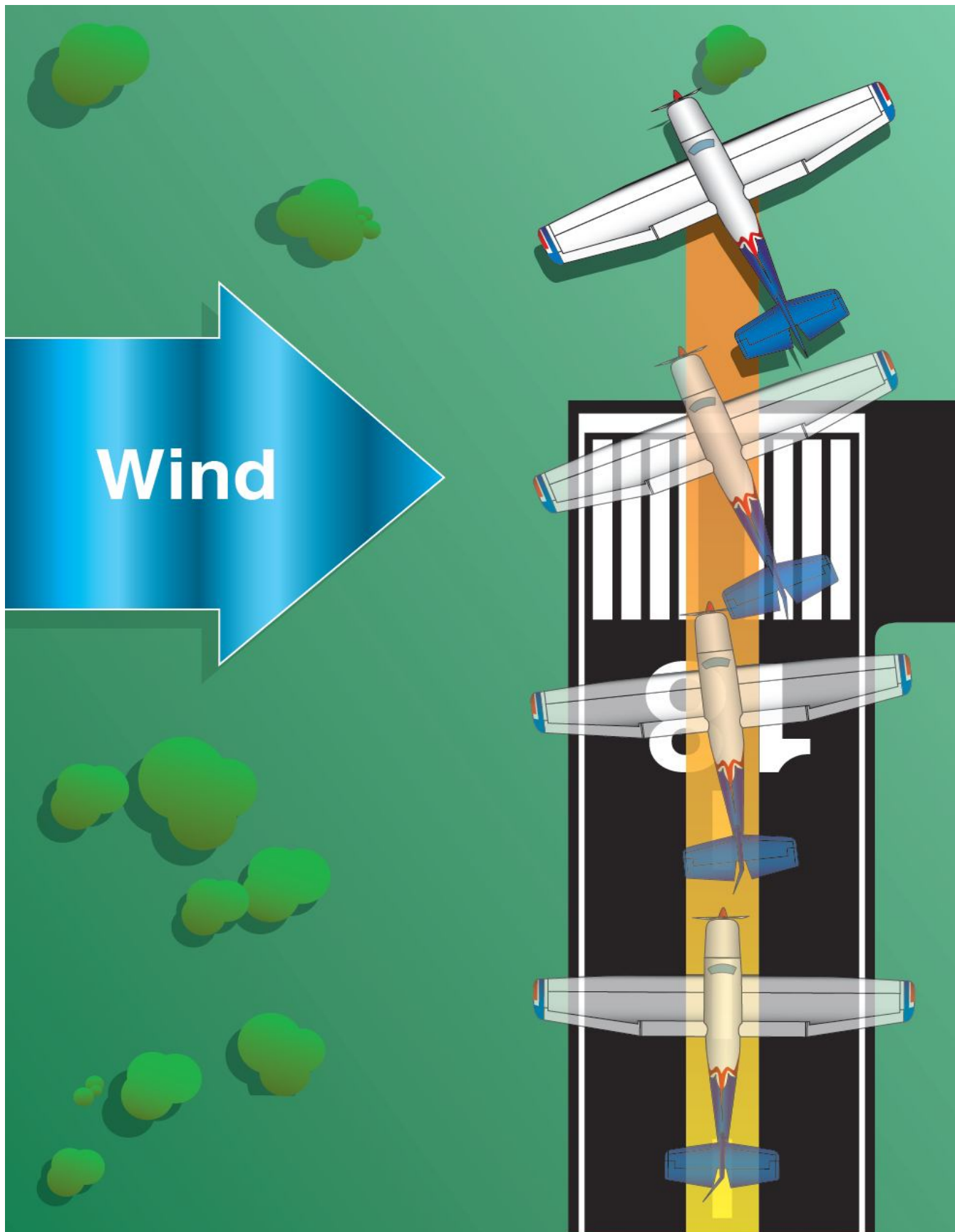


Figure 6-6. Crosswind climb flightpath.

The most common errors made while performing crosswind takeoffs include the following:

- Failure to review AFM/POH performance and charts prior to takeoff.
- Failure to adequately clear the area prior to taxiing onto the active runway.
- Using less than full aileron pressure into the wind initially on the takeoff roll.
- Mechanical use of aileron control rather than judging lateral position of airplane on runway from visual clues and applying sufficient aileron to keep airplane centered laterally on runway.
- Side-skipping due to improper aileron application.
- Inadequate rudder control to maintain airplane parallel to centerline and pointed straight ahead in alignment with visual references.
- Excessive aileron input in the latter stage of the takeoff roll resulting in a steep bank into the wind at lift-off.
- Inadequate drift correction after lift-off.

Ground Effect on Takeoff

Ground effect is a condition of improved performance encountered when the airplane is operating very close to the ground. Ground effect can be detected and normally occurs up to an altitude equal to one wingspan above the surface. [Figure 6-7] Ground effect is most significant when the airplane maintains a constant attitude at low airspeed at low altitude (for example, during takeoff when the airplane lifts off and accelerates to climb speed, and during the landing flare before touchdown).

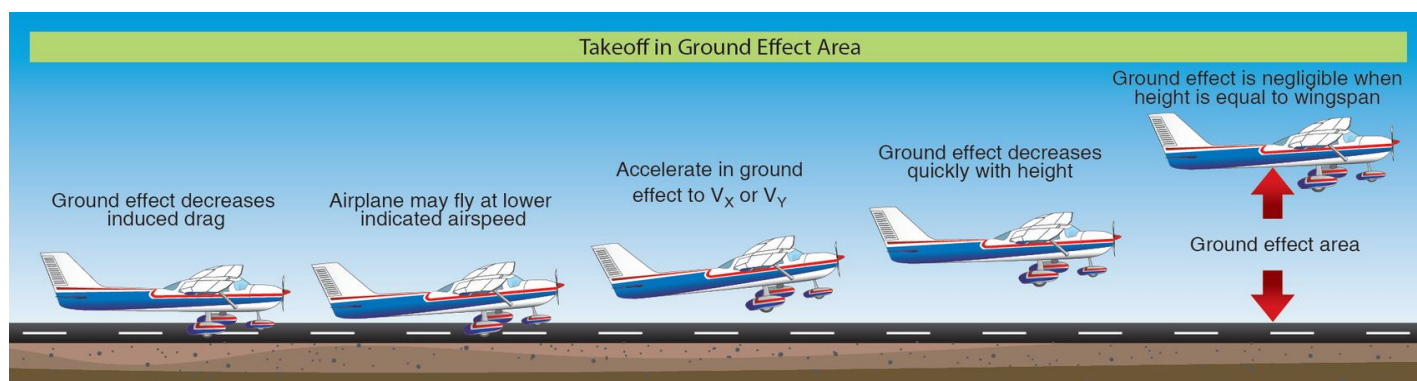


Figure 6-7. Takeoff in ground effect area.

When the wing is under the influence of ground effect, there is a reduction in upwash, downwash, and wingtip vortices. As a result of the reduced wingtip vortices, induced drag is reduced. When the wing is at a height equal to 1/4 the span, the reduction in induced drag is about 25 percent. When the wing is at a height equal to 1/10 the span, the reduction in induced drag is about 50 percent. At high speeds where parasite drag dominates, induced drag is a small part of the total drag. Consequently, ground effect is a greater concern during takeoff and landing.

At takeoff, the takeoff roll, lift-off, and the beginning of the initial climb are accomplished within the ground effect area. The ground effect causes local increases in static pressure, which cause the airspeed indicator and altimeter to indicate slightly lower values than they should and usually cause the vertical speed indicator to indicate a descent. As the airplane lifts off and climbs out of the ground effect area, the following occurs:

- The airplane requires an increase in AOA to maintain lift coefficient.
- The airplane experiences an increase in induced drag and thrust required.
- The airplane experiences a pitch-up tendency and requires less elevator travel because of an increase in downwash at the horizontal tail.
- The airplane experiences a reduction in static source pressure and a corresponding increase in indicated airspeed.

V_X is the speed at which the airplane achieves the greatest gain in altitude for a given distance over the ground. It is usually slightly less than V_Y , which is the greatest gain in altitude per unit of time. The specific speeds to be used for a given airplane are stated in the FAA-approved AFM/POH. The pilot should be aware that, in some airplanes, a deviation of 5 knots from the recommended speed may result in a significant reduction in climb performance; therefore, the pilot should maintain precise control of the airspeed to ensure the maneuver is executed safely and successfully.

Due to the reduced drag in ground effect, the airplane may seem to be able to take off below the recommended airspeed. However, as the airplane climbs out of ground effect below the recommended climb speed, initial climb performance will be much less than at V_Y or even V_X . Under conditions of high density altitude, high temperature, and/or maximum gross weight, the airplane may be able to lift off but will be unable to climb out of ground effect. Consequently, the airplane may not be able to clear obstructions. Lift-off before attaining recommended flight airspeed incurs more drag, which requires more power to overcome. Since the initial takeoff and climb is based on maximum power, reducing drag is the only option. To reduce drag, pitch should be reduced which means losing altitude. Pilots should remember that many airplanes cannot safely takeoff at maximum gross weight at certain altitudes and temperatures, due to lack of performance. Therefore, under marginal conditions, it is important that the airplane takes off at the speed recommended for adequate initial climb performance.

Ground effect is important to normal flight operations. If the runway is long enough or if no obstacles exist, ground effect can be used to the pilot's advantage by using the reduced drag to improve initial acceleration.

When taking off from an unsatisfactory surface, the pilot should apply as much weight to the wings as possible during the ground run and lift-off, using ground effect as an aid, prior to attaining true flying speed. The pilot should reduce AOA to attain normal airspeed before attempting to fly out of the ground effect areas.

Short-Field Takeoff and Maximum Performance Climb

When performing takeoffs and climbs from fields where the takeoff area is short or the available takeoff area is restricted by obstructions, the pilot should operate the airplane at the maximum limit of its takeoff performance capabilities. To depart from such an area safely, the pilot needs to exercise positive and precise control of airplane attitude and airspeed, so that takeoff and climb performance result in the shortest ground roll and the steepest angle of climb. [Figure 6-8] The pilot should consult and follow the performance section of the AFM/POH to obtain the power setting, flap setting, airspeed, and procedures prescribed by the airplane's manufacturer.

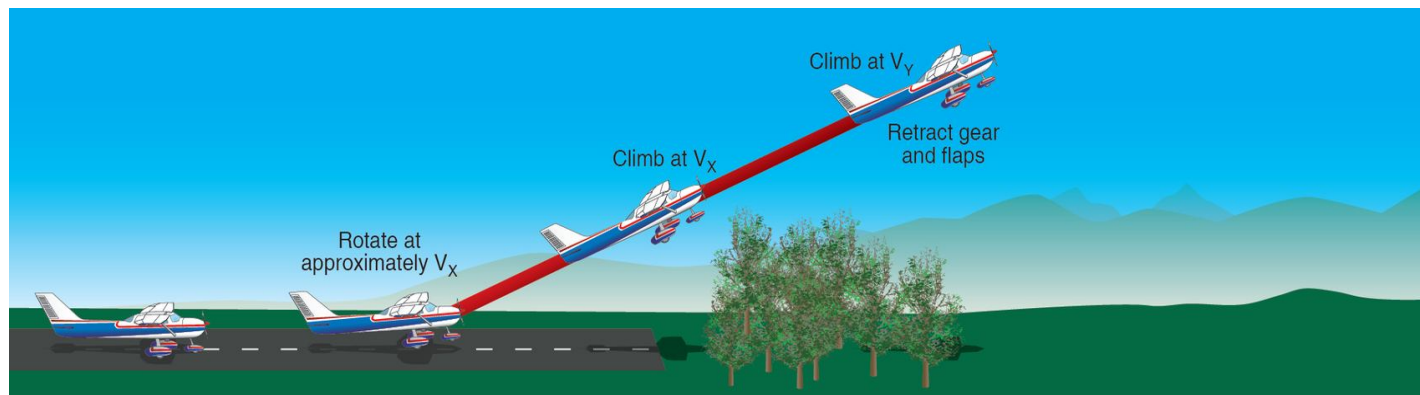


Figure 6-8. Short-field takeoff.

The pilot should have adequate knowledge in the use and effectiveness of the best angle-of-climb speed (V_X) and the best rate-of-climb speed (V_Y) for the specific make and model of airplane being flown in order to safely accomplish a takeoff at maximum performance.

Takeoff Roll

Taking off from a short field requires the takeoff to be started from the very beginning of the takeoff area. At this point, the airplane is aligned with the intended takeoff path. If the airplane manufacturer recommends the use of flaps, they are extended the proper amount before beginning the takeoff roll. This allows the pilot to devote full attention to the proper technique and the airplane's performance throughout the takeoff.

The pilot should apply takeoff power smoothly and continuously, without hesitation, to accelerate the airplane as rapidly as possible. Some pilots prefer to hold the brakes until the maximum obtainable engine revolutions per minute (rpm) are achieved before allowing the airplane to begin its takeoff run. However, it has not been established that this procedure results in a shorter takeoff run in all light, single-engine airplanes. The airplane is allowed to roll with its full weight on the main wheels and accelerate to the lift-off speed. As the takeoff roll progresses, the pilot should adjust the airplane's pitch attitude and AOA to attain minimum drag and maximum acceleration. In nose-wheel type airplanes, this involves little use of the elevator control since the airplane is already in a low-drag attitude.

Lift-Off

As V_X approaches, the pilot should apply back-elevator pressure until reaching the appropriate V_X attitude to ensure a smooth and firm lift-off, or rotation. Since the airplane accelerates more rapidly after lift-off, the pilot should apply additional back-elevator pressure to hold a constant airspeed. After becoming airborne, the pilot will maintain a wings-level climb at V_X until all obstacles have been cleared, or if no obstacles are present, until reaching an altitude of at least 50 feet above the takeoff surface. Thereafter, the pilot may lower the pitch attitude slightly and continue the climb at V_Y until reaching a safe maneuvering altitude. The pilot should always remember that an attempt to pull the airplane off the ground prematurely, or to climb too steeply, may cause the airplane to settle back to the runway or make contact with obstacles. Even if the airplane remains airborne, until the pilot reaches V_X , the initial climb will remain flat, which diminishes the pilot's ability to successfully perform the climb and/or clear obstacles. [Figure 6-9]

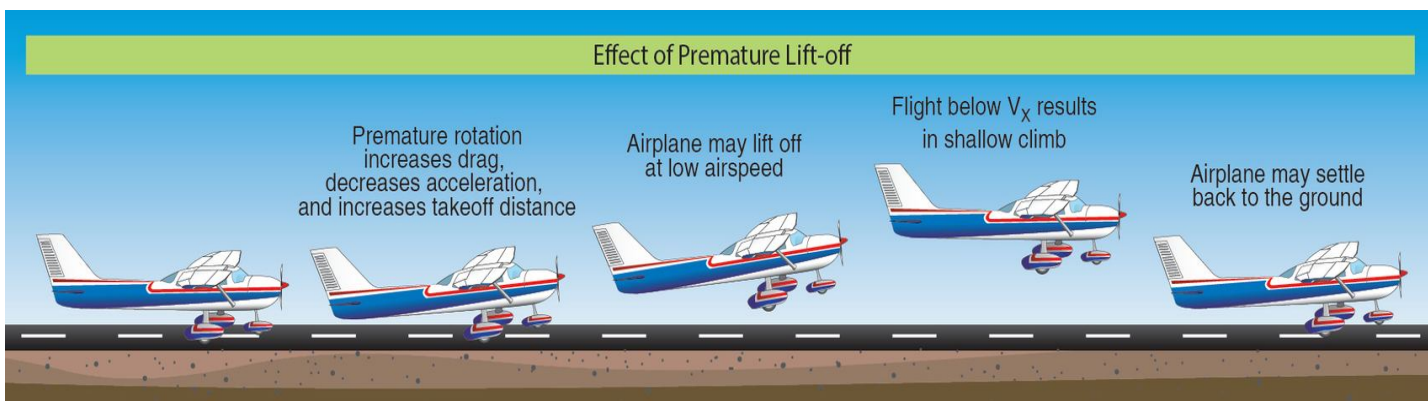


Figure 6-9. Effect of premature lift-off.

The objective is to rotate to the appropriate pitch attitude at (or near) V_X . The pilot should be aware that some airplanes have a natural tendency to lift off well before reaching V_X . In these airplanes, it may be necessary to allow the airplane to lift off in ground effect and then reduce pitch attitude to level until the airplane accelerates to V_X with the wheels just clear of the runway surface. This method is preferable to forcing the airplane to remain on the ground with forward elevator-control pressure until V_X is attained. Holding the airplane on the ground unnecessarily puts excessive pressure on the nose-wheel and may result in "wheel barrowing." It also hinders both acceleration and overall airplane performance.

Initial Climb

On short-field takeoffs, the landing gear and flaps should remain in takeoff position until the airplane is clear of obstacles (or as recommended by the manufacturer) and V_Y has been established. Until all obstacles have been cleared, the pilot should maintain focus outside the airplane instead of reaching for landing gear or flap controls or looking inside the airplane for any reason. When the airplane is stabilized at V_Y , the landing gear (if retractable) and flaps should be retracted. It is usually advisable to raise the flaps in increments to avoid sudden loss of lift and settling of the airplane. The pilot should next reduce the power to the normal climb setting or as recommended by the airplane manufacturer.

Common errors in the performance of short-field takeoffs and maximum performance climbs are:

- Failure to review AFM/POH and performance charts prior to takeoff.
- Failure to adequately clear the area.
- Failure to utilize all available runway/takeoff area.
- Failure to have the airplane properly trimmed prior to takeoff.
- Premature lift-off resulting in high drag.
- Holding the airplane on the ground unnecessarily with excessive forward-elevator pressure.
- Inadequate rotation resulting in excessive speed after lift-off.
- Inability to attain/maintain V_X .
- Fixation on the airspeed indicator during initial climb.
- Premature retraction of landing gear and/or wing flaps.

Soft/Rough-Field Takeoff and Climb

Takeoffs and climbs from soft fields require the use of operational techniques for getting the airplane airborne as quickly as possible to eliminate the drag caused by tall grass, soft sand, mud, and snow and may require climbing over an obstacle. The technique makes judicious use of ground effect to reduce landing gear drag and requires an understanding of the airplane's slow speed characteristics and responses. These same techniques are also useful on a rough field where the pilot should get the airplane off the ground as soon as possible to avoid damaging the landing gear.

Taking off from a soft surface or through soft surfaces or long, wet grass reduces the airplane's ability to accelerate during the takeoff roll and may prevent the airplane from reaching adequate takeoff speed if the pilot applies normal takeoff techniques. The pilot should be aware that the correct takeoff procedure for soft fields is quite different from the takeoff procedures used for short fields with firm, smooth surfaces. To minimize the hazards associated with takeoffs from soft or rough fields, the pilot should transfer the support of the airplane's weight as rapidly as possible from the wheels to the wings as the takeoff roll proceeds by establishing and maintaining a relatively high AOA or nose-high pitch attitude as early as possible. The pilot should lower the wing flaps prior to starting the takeoff (if recommended by the manufacturer) to provide additional lift and to transfer the airplane's weight from the wheels to the wings as early as possible. The pilot should maintain a continuous motion with sufficient power while lining up for the takeoff roll as stopping on a soft surface, such as mud or snow, might bog the airplane down.

Takeoff Roll

As the airplane is aligned with the takeoff path, the pilot should apply takeoff power smoothly and as rapidly as the powerplant can accept without faltering. As the airplane accelerates, the pilot should apply enough back-elevator pressure to establish a positive AOA and to reduce the weight supported by the nose-wheel.

When the airplane is held at a nose-high attitude throughout the takeoff run, the wings increasingly relieve the wheels of the airplane's weight as speed increases and lift develops, thereby minimizing the drag caused by surface irregularities or adhesion. If this attitude is accurately maintained, the airplane virtually flies itself off the ground, becoming airborne but at an airspeed slower than a safe climb speed because of ground effect. [Figure 6-10]



Figure 6-10. Soft-field takeoff.

Lift-Off

After the airplane becomes airborne, the pilot should gently lower the nose with the wheels clear of the surface to allow the airplane to accelerate to a minimum safe climb out speed. Immediately after the airplane becomes airborne and while it accelerates, the pilot should be aware that, while transitioning out of the ground effect area, the airplane will have a tendency to settle back onto the surface, even with full power applied. Therefore, it is essential that the airplane remain in ground effect until at least V_X is reached. This requires a good understanding of the control pressures, aircraft responses, visual clues, and acceleration characteristics of that particular airplane.

Initial Climb

After a positive rate of climb is established, and the airplane has accelerated to V_Y , the pilot should retract the landing gear and flaps, if equipped. If departing from an airstrip with wet snow or slush on the takeoff surface, the gear should not be retracted immediately so that any wet snow or slush can be air-dried. In the event an obstacle needs to be cleared after a soft-field takeoff, the pilot should perform the climb-out at V_X until the obstacle has been cleared. The pilot should then adjust the pitch attitude to V_Y and retract the gear and flaps. The power can then be reduced to the normal climb setting.

Common errors in the performance of soft/rough field takeoff and climbs are:

- Failure to review AFM/POH and performance charts prior to takeoff.
- Failure to adequately clear the area.
- Insufficient back-elevator pressure during initial takeoff roll resulting in inadequate AOA.
- Failure to cross-check engine instruments for indications of proper operation after applying power.
- Poor directional control.
- Climbing too high after lift-off and not leveling off low enough to maintain ground effect attitude.
- Abrupt and/or excessive elevator control while attempting to level off and accelerate after liftoff.
- Allowing the airplane to "mush" or settle resulting in an inadvertent touchdown after lift-off.
- Attempting to climb out of ground effect area before attaining sufficient climb speed.
- Failure to anticipate an increase in pitch attitude as the airplane climbs out of ground effect.

Rejected Takeoff/Engine Failure

Emergency or abnormal situations can occur during a takeoff that require a pilot to reject the takeoff while still on the runway. Circumstances such as a malfunctioning powerplant, inadequate acceleration, runway incursion, or air traffic conflict may be reasons for a rejected takeoff.

Prior to takeoff, the pilot should identify a point along the runway at which the airplane should be airborne. If that point is reached and the airplane is not airborne, immediate action should be taken to discontinue the takeoff. When properly planned and executed, the airplane can be stopped on the remaining runway without using extraordinary measures, such as excessive braking that may result in loss of directional control, airplane damage, and/or personal injury. The POH/AFM ground roll distances for take-off and landing added together provide a good estimate of the total runway needed to accelerate and then stop.

In the event a takeoff is rejected, the power is reduced to idle and maximum braking applied while maintaining directional control. If it is necessary to shut down the engine due to a fire, the mixture control should be brought to the idle cutoff position and the magnetos turned off. In all cases, the manufacturer's emergency procedure should be followed.

Urgency characterizes all power loss or engine failure occurrences after lift-off. In most instances, the pilot has only a few seconds after an engine failure to decide what course of action to take and to execute it.

In the event of an engine failure on initial climb-out, the pilot's first responsibility is to maintain aircraft control. At a climb pitch attitude without power, the airplane is at or near a stalling AOA. At the same time, the pilot may still be holding right rudder. The pilot should immediately lower the nose to prevent a stall while moving the rudder to ensure coordinated flight. The pilot should establish a controlled glide toward a plausible landing area, preferably straight ahead. Attempting to turn back to the takeoff runway should not be attempted unless the pilot previously trained for an emergency turn-back and sufficient altitude exists.

Noise Abatement

Aircraft noise problems are a major concern at many airports throughout the country. Many local communities have pressured airports into developing specific operational procedures that help limit aircraft noise while operating over nearby areas. As a result, noise abatement procedures have been developed for many of these airports that include standardized profiles and procedures to achieve these lower noise goals.

Airports that have noise abatement procedures provide information to pilots, operators, air carriers, air traffic facilities, and other special groups that are applicable to their airport. These procedures are available to the aviation community by various means. Most of this information comes from the Chart Supplements, local and regional publications, printed handouts, operator bulletin boards, safety briefings, and local air traffic facilities.

At airports that use noise abatement procedures, reminder signs may be installed at the taxiway hold positions for applicable runways to remind pilots to use and comply with noise abatement procedures on departure. Pilots who are unfamiliar with these procedures should ask the tower or air traffic facility for the recommended procedures. In any case, pilots should be considerate of the surrounding community while operating their airplane to and from such an airport. This includes operating as quietly, and safely as possible.

Chapter Summary

The takeoff and initial climb are relatively short phases required for every flight and are often taken for granted, yet 1 out of 5 accidents occur during this phase and half the mishaps are the result of pilot error. Becoming proficient in and applying the techniques and principles discussed in this chapter help pilots reduce their susceptibility to becoming a mishap statistic.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 7: Ground Reference Maneuvers

Introduction

During initial training, pilots learn how various flight control pressure inputs affect the airplane. After achieving a sufficient level of competence, the pilot is ready to apply this skill and maintain the airplane, not only at the correct attitude and power configuration, but also along an appropriate course relative to objects on the ground. This skill is the basis for traffic patterns, survey, photographic, sight-seeing, aerial application (crop dusting), and various other flight profiles requiring specific flightpaths referenced to points on the surface.

Ground reference maneuvers are the principal flight maneuvers that combine the four fundamentals (straight-and-level, turns, climbs, and descents) into a set of integrated skills that the pilot uses in everyday flight activity. From every takeoff to every landing, a pilot exercises these skills to control the airplane. Therefore, a pilot needs to develop the proper coordination, timing, and attention in order to accurately and safely maneuver the airplane with regard to the required attitudes and ground references.

The pilot should be introduced by their instructor to ground reference maneuvers as soon as the pilot shows proficiency in the four fundamentals. Ground reference maneuvers call for manipulation of the flight controls using necessary control pressures to affect the airplane's attitude and position by using the outside natural horizon and ground-based references with brief periods of scanning the flight instruments.

Maneuvering by Reference to Ground Objects

Ground reference maneuvers train the pilot to accurately place the airplane in relationship to specific references and maintain a desired ground track. While vision is the most utilized sense, other senses are actively involved at different levels. For example, the amount of pressure needed to overcome flight control surface forces provides tactile feedback as to the airplane's airspeed and aerodynamic load.

It is a common error for beginning pilots to fixate on a specific reference, such as a single location on the ground or a spot on the natural horizon. A pilot fixating on any one reference loses the ability to determine rate, which significantly degrades a pilot's performance. By visually scanning across several references, the pilot learns how to determine the rate of closure to a specific point. In addition, the pilot should scan between several visual references to determine relative motion and to determine if the airplane is maintaining, or drifting to or from, the desired ground track. Consider a skilled automobile driver in a simple intersection turn; the driver does not merely turn the steering wheel some degree and hope that it will work out. The driver picks out several references, such as an island to their side, a painted lane line, or the opposing curb, and uses those references to make almost imperceptible adjustments to the amount of deflection on the steering wheel. At the same time, the driver adjusts the pressure on the accelerator pedal to smoothly join the new lane. In the same manner, multiple references are required to precisely control the airplane in reference to the ground.

Not all ground-based references are visually equal. Awareness of typical visual illusions helps a pilot select appropriate references. For example, larger objects or references may appear closer than they actually are when compared to smaller objects or references. Prevailing visibility has a significant effect on the pilot's perception of the distance to a reference. Excellent visibility with clear skies tends to make an object or reference appear closer than when compared to a hazy day with poor visibility. Rain can alter the visual image in a manner creating an illusion of being at a higher than actual altitude, and brighter objects or references may appear closer than dimmer objects. However, if using references of similar size and proportion, pilots find ground reference maneuvers easier to execute.

Ground-based references can be numerous. Examples include breakwaters, canals, fence lines, field boundaries, highways, railroad tracks, roads, pipe lines, power lines, water tanks, and many other objects; however, choices can be limited by geography, population density, infrastructure, or structures. The pilot should consider the type of maneuver being performed, altitude at which the maneuver will be performed, emergency landing requirements, density of structures, wind direction, visibility, and the type of airspace when selecting a ground-based reference.

Ground reference maneuvers develop a pilot's division of attention skill. A pilot needs to control the airplane's attitude while tracking a specific path over the ground. In addition, the pilot should be able to scan for hazards such as other aircraft, prepare for an emergency landing should the need arise, and scan the flight and engine instruments at regular intervals to ensure that a pending situation, such as decreasing oil pressure, does not turn into an unexpected incident.

Ground reference maneuvers place the airplane in a low altitude environment with associated hazards. Pilots should look for other aircraft, including helicopters, and look for obstructions such as radio towers and wires. In addition, pilots should consider engine failure and have one or more locations available for an emergency landing. Pilots should always clear the area with two 90° clearing turns looking to the left and the right, as well as above and below the airplane. The maneuver area should not cause disturbances and be well away from any open air assembly of persons, congested areas of a city, town, or settlement, or herd of livestock. Before performing any maneuver, the pilot should complete the required checklist items, make any radio announcements (such as on a practice area frequency), and safety clearing turns. As a general note, a ground reference maneuver should not exceed a bank angle of 45° or an airspeed greater than the maneuvering speed. As part of preflight planning, the pilot should determine the predicted (POH/AFM) stall speed at 50° or at the highest bank angle expected during the maneuver to assure there will be a safety margin above the stall speed during the maneuver.

Drift and Ground Track Control

Wind direction and velocity variations create the need for flightpath corrections during a ground reference maneuver. In a similar way that water currents affect the progress of a boat or ship, wind directly influences the path that the airplane travels in reference to the ground. Whenever the airplane is in flight, the movement of the air directly affects the actual ground track of the airplane.

For example, an airplane is traveling at 90 knots (90 nautical miles per hour) and the wind is blowing from right to left at 10 knots. The airplane continues forward at 90 knots but also travels left 10 nautical miles for every hour of flight time. If the airplane, in this example doubles its speed to 180 knots, it still drifts laterally to the left 10 nautical miles every hour. Unless in still air, traveling to a point on the surface requires compensation for the movement of the air mass.

Ground reference maneuvers are generally flown at altitudes between 600 and 1,000 feet above ground level (AGL). The pilot should consider the following when selecting the maneuvering altitude:

- The lower the maneuvering altitude, the faster the airplane appears to travel in relation to the ground.
- Drift should be easily recognizable from both sides of the airplane.
- The altitude should provide obstruction clearance of no less than 500 feet vertically above the obstruction and 2,000 feet horizontally.
- In the event of an engine failure, lower altitudes equate to less time to configure the airplane and reduced gliding distance before a forced landing.
- What specific altitude or altitude range does the testing standard call for?

Correcting Drift During Straight-and-Level Flight

When flying straight and level and following a selected straight-line direct ground track, the preferred method of correcting for wind drift is to angle the airplane sufficiently into the wind to cancel the effect of the sideways drift caused by the wind. The wind's speed, the angle between the wind direction and the airplane's longitudinal axis, and the airspeed of the airplane determine the required wind correction angle. For example, an airplane with an airspeed of 100 knots in an air mass moving at 20 knots directly from the side, should turn 12° into the wind to cancel the airplane's drift. If the wind in the above example is only 10 knots, the wind correction angle required to cancel the drift is six degrees. When the drift has been neutralized by heading the airplane into the wind, the airplane will fly the direct straight ground track.

To further illustrate this point, if a boat is crossing a river and the river's current is completely still, the boat could head directly to a point on the opposite shore on a straight course without any drift. However, rivers tend to have a downstream current that needs to be considered if the captain wants the boat to arrive at the opposite shore using a direct straight path. Any downstream current pushes the boat sideways and downstream at the speed of the current. To counteract this downstream movement, the boat needs to move upstream at the same speed as the river is moving the boat downstream. This is accomplished by angling the boat upstream to counteract the downstream flow. If done correctly, the boat follows a direct straight track across the river to the intended destination point. A slower forward speed of the boat or a faster river current requires a greater angle to counteract the drift. *[Figure 7-1]*



Figure 7-1. Wind drift.

As soon as the pilot lifts off the surface and levels the wings in a crosswind, the airplane begins tracking sideways. The force of the crosswind acts on the mass of the airplane, and the speed of drift increases up to the speed of the crosswind component. A wind that is directly to the right or the left (at a 90° angle) will cause the airplane to accelerate sideways at the same speed as the wind. When the wind is halfway between the side and the nose of the airplane (at a 45° angle), it causes a sideways drift up to just over 70 percent of the total speed of the wind. It should be understood that pilots do not calculate the required drift correction angles for ground reference maneuvers; they merely use the references and adjust the airplane's relationship to those references to cancel any drift. The groundspeed of the airplane is also affected by the wind. As the wind direction becomes parallel to the airplane's longitudinal axis, the magnitude of the wind's effect on the groundspeed is greater; as the wind becomes perpendicular to the longitudinal axis, the magnitude of the wind's effect on the groundspeed is less. In general, When the wind is blowing straight into the nose of the airplane, the groundspeed will be less than the airspeed. When the wind is blowing from directly behind the airplane, the groundspeed will be faster than the airspeed. In other words, when the airplane is headed upwind, the groundspeed is decreased; when headed downwind, the groundspeed is increased.

Constant Radius During Turning Flight

In a no-wind condition, a pilot may make a constant-radius turn over the ground using a fixed bank angle. If wind is present, however, a pilot will observe a change in the radius of a turn while maintaining that same constant bank angle. [Figure 7-2] As groundspeed increases, the observed radius of the turn increases. Conversely, as groundspeed decreases, the radius of the turn over the ground will decrease. For a ground-referenced constant-radius turn, the pilot compensates for changes in groundspeed by varying the bank angle throughout the turn. When groundspeed increases, the pilot banks more steeply to maintain a constant-radius turn over the ground. The converse is also true: when groundspeed decreases, the pilot uses a shallower bank.



Figure 7-2. *Effect of wind during a turn.*

For a given true airspeed, the radius of turn in the air varies proportionally with the bank angle. To maintain a constant radius over the ground, the bank angle used is proportional to groundspeed. For example, an airplane is in the downwind position at 100 knots groundspeed. In this example, the wind is 10 knots, meaning that the airplane has an airspeed of 90 knots (for this discussion, assume true, calibrated, and indicated airspeed are all the same). If the pilot starts a turn using a 45° bank angle, the turn radius over the ground at that moment is approximately 890 feet. As the airplane turns, the groundspeed decreases and the bank angle needs to be reduced in order to maintain the same turn radius of 890 feet over the ground. At the upwind point of the turn, the bank angle should be approximately 33° . In another example, if the downwind is flown at an airspeed of 90 knots in a 10 knot tailwind with a desired turn radius of 2,000 feet, the bank angle would be approximately 24° . The bank angle flying upwind would be approximately 16° .

Put another way, at a higher groundspeed, there is less time to turn the airplane while trying to maintain a ground-referenced constant-radius turn. The pilot increases the bank angle in order to increase the rate of turn, and the increased rate of turn offsets the reduced time available to make the turn. Conversely, when flying at a lower groundspeed, the pilot reduces the angle of bank and rate of turn to compensate for the additional time taken while making the turn. With some experience, pilots may notice how wind direction affects the time needed for various segments of ground-referenced turns.

To demonstrate the effect that wind has on turns, the pilot should select a straight-line ground reference, such as a road or railroad track. [Figure 7-3] Choosing a straight-line ground reference that is parallel to the wind, the airplane would be flown into the wind and directly over the selected straight-line ground reference. Once a straight-line ground reference is established, the pilot makes a 360° constant medium-banked turn. As the airplane completes the 360° turn, it should return directly over the straight-line ground reference but downwind from the starting point. Choosing a straight-line ground reference that has a crosswind, and using the same 360° constant medium-banked turn, demonstrates how the airplane drifts away from the reference even as the pilot holds a constant bank angle. In both examples, the path over the ground is not circular, although in reference to the air, the airplane flew a perfect continuous radius.



Figure 7-3. *Effect of wind during turn.*